

Transaction-level approval enforcement on Besu

Proof of concept, 15 September 2026 · unmodified Besu 26.8.1 — the build Lineth pins · measured on one laptop (Apple M2 Max)

It works as a plain Besu plugin — no fork, no patch, one JAR. The node refuses to include any transaction whose actual execution differs from what was authorized: the block producer compares a fingerprint of the execution it just performed against a signed approval and drops the transaction *before* the state change is committed. Not "it fails with a receipt" — it never enters the block. **Throughput cost: about 7 %.** The remaining ceiling belongs to Besu itself, not to the gate.

WHAT WAS RUN ON WHAT

The node is Besu, and Lineth is stock Besu plus its own plugin JARs — so the plugin was built and measured against the same upstream Besu 26.8.1 that Lineth pins, with nothing patched. **Lineth's own sequencer and transaction-pool plugins were loaded alongside ours in the functional suite**, where both plugins and both tracers run in one node and our gate still decides. The throughput figures below were measured **without** Lineth's plugins loaded, so they do not include the cost of its ZK line-counting tracer — that remains unmeasured.

MEASURED THROUGHPUT — 60 S ADAPTIVE LOAD, ETH TRANSFERS, 10 WALLETS, 1 S BLOCKS

Configuration	Avg tx/s	Confirmed	Failed	Block fill
Straight to Besu — neither the proxy nor the gate in the path	765	45,887	0	15.9 %
Through the authorization proxy, gate off	727	43,622	0	—
Through the authorization proxy, gate on	671	40,198	0	—

Read it this way: the gate costs ~7 % and the proxy's own authorization work ~5 %. Taking both out entirely buys only 13 %, and at that ceiling Besu's blocks are just **15.9 % full** — so what runs out first is Besu's per-transaction RPC and pool handling, not block space or our enforcement. Zero failures in every run; all 40,198 confirmations were re-verified against on-chain receipts.

WHERE THE TIME GOES

- Inside the block producer: 7–22 µs per transaction** for the plugin's own work (approval lookup, fingerprint, comparison). Negligible.
- Per client request: 17.5 ms → 21.1 ms** end-to-end through the proxy. The ~3.5 ms the gate adds is a pre-flight simulation on the node plus a signature.
- Confirmation latency ~1.0 s median in every configuration — that is the 1 s block cadence, unchanged by the gate.

WHAT WAS VERIFIED

- 21 end-to-end scenarios / 23 checks green on a real node, including **contract deployment**.
- Coexists with Lineth's own sequencer and pool plugins** — their published JARs, both plugins and both tracers in one node; our gate still decides.
- The headline case: an approved call overtaken in the same block by a higher-fee transaction that redirects it is **excluded** — no receipt, nonce and balances untouched.

LIMITS WORTH KNOWING BEFORE ANYONE PLANS AROUND THIS

- Enforcement is producer-only.** It is a block-building rule, not a consensus rule: every producer must run the plugin, and a block received from elsewhere is not re-checked. Making it a consensus rule would require changing Besu, not a plugin.
- Deployments and other state-binding transactions are approved against exact state.** A concurrent transaction in the same block touching one of the same storage slots invalidates the approval; the client must re-submit after a fresh pre-flight. Ordinary calls use a looser binding and are unaffected.
- An expired approval is dropped at the next block-building round**, not on a timer — Besu gives plugins no way to remove a transaction from the pool. In the meantime it simply cannot be selected.
- Signing key rotation is now add-then-switch** — a batch names its key, the sequencer trusts a set; the seed reaches OPS like every other OPS secret (Secrets Manager via IRSA/CSI). Approvals still carry no expiry: a permission revoked after approval does not revoke the approval until the transaction leaves the pool.
- Version coupling.** The Besu plugin interfaces used here are marked unstable and were split into separate artifacts in 26.8.x. Pin the node version and re-run the suite on every upgrade. Requires JDK 25.
- Co-running with Lineth's ZK tracer requires an Osaka-fork chain;** on older forks that tracer refuses and block building stops. This is their constraint, not ours, but it pins the network's fork — and the throughput numbers above were taken without it running.
- Reorgs need an answer.** Transactions returned to the pool by a reorg wait for a fresh approval, which the proxy does not currently re-send. Approvals also cannot be revoked once issued (they expire).
- Transaction types are a PoC scope limit, not a technical barrier.** The gate approves protected legacy and EIP-1559; access-list (EIP-2930), blob (EIP-4844) and set-code (EIP-7702) transactions are refused rather than approved. **OPS itself has no such limit** — it forwards every type; this is the approval path only. Access-list and blob support is hours of work, since neither changes execution semantics. **EIP-7702 is doable but is real work:** its authorization list installs code on an EOA outside the call tree, so the signed envelope has to carry it — a day or two across both encoders, plus a decision on what a delegation means for contract ownership. Nothing here is blocked; it was out of scope. Also unchecked: bundle-style plugins that commit transactions themselves.
- All numbers come from one laptop** running the load generator, the proxy, PostgreSQL, Redis and Besu together. They bound this machine, not the design — a production sizing exercise is still owed.

Evidence: dashboard screenshots, per-run history records, status samples and gzipped receipt lists for all three configurations, plus the scenario suite and the reproduction commands, live with the PoC branch (poc/ops-besu-signed-approvals, see DEMO.md). Nothing here is simulated: the runs use the real authorization service, its own database and signing keys, and a real Besu node.